WOA — resultados CEC

Benchmark MAREA · paper OE1 · 71 funciones CEC2014/2017/2022 · 51 corridas por funcion

Protocolo

Valores = **error al optimo**, f(x) - f*. Menor es mejor; 0 es el optimo global. Dominio [-100,100] en todas las funciones. Poblacion N=80, iteraciones T=MaxFES/N. Presupuestos 5.000 / 50.000 / 500.000 evaluaciones. Las 30 funciones de CEC2014 y las 29 de CEC2017 corren en D=10 y D=50; las 12 de CEC2022 solo llegan a D=20, asi que en la columna D50 corren en **D=20** (marcado en la tabla). Todos los resultados vienen del mismo entorno del Mac mini (python 3.11.9, numpy 2.3.3, opfunu 1.0.4).

Resumen — posicion entre los 11 algoritmos

dimension	presupuesto	rango medio de Friedman	posicion de 11	funciones donde es el mejor
D10	5k FES	3.34	2º	4 de 71
D10	50k FES	4.55	4º	0 de 71
D10	500k FES	5.37	4º	1 de 71
D50	5k FES	3.79	3º	5 de 71
D50	50k FES	4.41	3º	2 de 71
D50	500k FES	6.06	7º	0 de 71

WOA — estadisticas por funcion, D = 10

error al optimo sobre 51 corridas · b=best w=worst med=median avg=mean sd=std

func	suite	dim	5k					50k					500k				
			b	w	med	avg	sd	b	w	med	avg	sd	b	w	med	avg	sd
F12014	2014	10	4.59e+05	1.80e+07	3.52e+06	4.94e+06	4.25e+06	5.77e+03	3.44e+06	2.15e+05	2.72e+05	4.61e+05	2.85e+04	2.62e+05	1.61e+05	1.45e+05	7.21e+04
F22014	2014	10	3.96e+05	5.55e+06	1.49e+06	1.79e+06	9.95e+05	5.19e+03	3.67e+04	1.49e+04	1.47e+04	6.28e+03	5.15e+01	1.19e+04	9.82e+03	7.59e+03	3.97e+03
F32014	2014	10	8.26e+02	4.67e+04	1.64e+04	1.71e+04	1.19e+04	5.40e+01	2.80e+03	6.87e+02	9.63e+02	7.81e+02	1.89e-01	9.57e+02	7.15e+00	1.96e+02	3.35e+02
F42014	2014	10	5.39e-01	1.16e+02	3.51e+01	3.30e+01	2.22e+01	2.96e-01	1.02e+02	3.48e+01	2.38e+01	1.92e+01	2.46e-01	3.48e+01	3.48e+01	1.94e+01	1.65e+01
F52014	2014	10	2.02e+01	2.07e+01	2.04e+01	2.04e+01	1.32e-01	2.00e+01	2.02e+01	2.01e+01	2.01e+01	4.46e-02	2.00e+01	2.02e+01	2.00e+01	2.00e+01	3.98e-02
F62014	2014	10	1.06e+00	1.13e+01	4.76e+00	4.76e+00	2.00e+00	2.75e-01	9.12e+00	4.01e+00	3.97e+00	1.68e+00	3.34e-01	7.58e+00	3.75e+00	3.81e+00	1.64e+00
F72014	2014	10	7.46e-01	2.31e+00	1.02e+00	1.04e+00	2.67e-01	7.34e-02	7.13e-01	2.48e-01	2.67e-01	1.12e-01	7.08e-04	4.98e-01	1.83e-01	1.95e-01	1.15e-01
F82014	2014	10	6.31e+00	4.36e+01	1.95e+01	2.02e+01	7.68e+00	1.00e+00	1.49e+01	5.98e+00	6.72e+00	3.13e+00	1.37e-05	1.99e+00	9.95e-01	6.63e-01	7.08e-01
F92014	2014	10	8.23e+00	5.37e+01	2.82e+01	3.10e+01	1.07e+01	4.03e+00	4.48e+01	2.59e+01	2.59e+01	8.86e+00	3.98e+00	2.89e+01	1.59e+01	1.60e+01	5.34e+00
F102014	2014	10	5.54e+01	7.45e+02	3.03e+02	3.31e+02	2.01e+02	3.13e+01	4.16e+02	1.91e+02	2.11e+02	1.11e+02	1.88e+01	2.96e+02	6.74e+01	1.06e+02	7.33e+01
F112014	2014	10	3.75e+02	1.58e+03	9.12e+02	9.32e+02	2.79e+02	2.45e+02	1.51e+03	6.65e+02	7.04e+02	2.80e+02	1.49e+02	1.08e+03	5.60e+02	5.50e+02	2.47e+02
F122014	2014	10	3.35e-01	1.33e+00	6.44e-01	7.00e-01	2.54e-01	8.92e-02	5.97e-01	2.60e-01	3.02e-01	1.46e-01	2.93e-02	6.10e-01	1.39e-01	1.65e-01	1.10e-01
F132014	2014	10	2.38e-01	8.62e-01	4.82e-01	4.88e-01	1.45e-01	1.13e-01	5.86e-01	2.96e-01	3.17e-01	1.05e-01	6.68e-02	3.82e-01	1.61e-01	1.66e-01	5.86e-02
F142014	2014	10	1.94e-01	1.08e+00	3.82e-01	5.26e-01	2.82e-01	1.04e-01	8.39e-01	2.75e-01	3.79e-01	2.15e-01	2.99e-02	2.56e-01	1.14e-01	1.15e-01	5.12e-02
F152014	2014	10	1.24e+00	5.15e+00	3.16e+00	3.16e+00	8.87e-01	4.74e-01	4.51e+00	1.41e+00	1.52e+00	7.18e-01	4.66e-01	2.31e+00	1.30e+00	1.32e+00	4.75e-01
F162014	2014	10	2.71e+00	4.01e+00	3.51e+00	3.46e+00	3.21e-01	1.78e+00	4.00e+00	3.02e+00	2.99e+00	4.19e-01	1.74e+00	3.52e+00	2.70e+00	2.67e+00	4.46e-01
F172014	2014	10	4.31e+03	7.14e+04	2.63e+04	2.73e+04	1.64e+04	2.38e+03	4.26e+04	1.49e+04	1.70e+04	9.80e+03	4.25e+02	1.52e+04	4.38e+03	4.94e+03	3.22e+03
F182014	2014	10	1.12e+04	1.07e+05	3.89e+04	4.30e+04	2.17e+04	2.29e+02	2.21e+04	6.77e+03	7.08e+03	5.05e+03	3.02e+01	3.24e+03	7.90e+02	1.09e+03	9.00e+02
F192014	2014	10	5.22e+00	1.79e+02	2.58e+01	3.96e+01	4.20e+01	3.67e+00	1.61e+02	1.52e+01	3.19e+01	4.04e+01	2.47e+00	2.68e+01	7.83e+00	8.21e+00	4.55e+00
F202014	2014	10	1.52e+04	9.22e+04	5.24e+04	5.17e+04	2.16e+04	4.54e+03	7.17e+04	3.28e+04	3.27e+04	1.67e+04	2.08e+02	2.92e+04	9.11e+03	1.18e+04	8.28e+03
F212014	2014	10	8.00e+02	5.25e+04	1.10e+04	1.49e+04	1.18e+04	4.11e+01	4.68e+03	7.49e+02	1.12e+03	1.06e+03	7.51e+00	1.25e+03	1.77e+02	2.34e+02	2.25e+02
F222014	2014	10	2.41e+01	8.73e+02	2.41e+02	1.84e+02	1.69e+02	2.07e+01	2.45e+02	2.72e+01	1.21e+02	1.09e+02	2.02e+01	2.41e+02	2.13e+01	5.97e+01	8.37e+01
F232014	2014	10	2.37e+02	2.42e+02	2.42e+02	2.41e+02	2.35e+00	3.49e-01	2.42e+02	2.37e+02	2.19e+02	3.68e+01	1.87e-02	2.37e+02	2.00e+02	2.05e+02	3.35e+01
F242014	2014	10	1.11e+02	1.68e+02	1.38e+02	1.37e+02	1.19e+01	1.13e+02	1.62e+02	1.34e+02	1.34e+02	1.13e+01	1.11e+02	1.44e+02	1.23e+02	1.24e+02	8.12e+00
F252014	2014	10	1.55e+02	2.06e+02	2.01e+02	2.01e+02	6.68e+00	1.47e+02	2.02e+02	2.01e+02	2.00e+02	7.58e+00	1.44e+02	2.02e+02	2.00e+02	1.99e+02	7.88e+00
F262014	2014	10	1.00e+02	1.01e+02	1.01e+02	1.01e+02	1.88e-01	1.00e+02	1.01e+02	1.00e+02	1.00e+02	1.31e-01	1.00e+02	1.00e+02	1.00e+02	1.00e+02	5.95e-02
F272014	2014	10	3.52e+02	5.32e+02	4.02e+02	4.08e+02	2.40e+01	3.56e+00	4.41e+02	4.00e+02	3.69e+02	1.07e+02	1.22e+00	4.05e+02	4.00e+02	3.15e+02	1.65e+02
F282014	2014	10	3.71e+02	7.09e+02	3.88e+02	4.44e+02	7.58e+01	3.69e+02	5.77e+02	3.81e+02	4.25e+02	5.86e+01	3.57e+02	5.09e+02	3.81e+02	4.19e+02	5.54e+01
F292014	2014	10	4.50e+06	8.25e+06	5.31e+06	5.40e+06	7.58e+05	4.50e+06	5.31e+06	5.31e+06	5.03e+06	3.85e+05	4.50e+06	5.31e+06	4.50e+06	4.69e+06	3.48e+05
F302014	2014	10	4.78e+03	3.94e+07	1.29e+05	2.10e+06	6.29e+06	3.06e+03	2.84e+06	4.90e+04	1.04e+05	3.95e+05	2.32e+03	1.17e+05	8.41e+03	2.79e+04	2.96e+04
F12017	2017	10	5.93e+05	1.84e+07	3.20e+06	3.77e+06	3.06e+06	4.89e+03	5.23e+04	2.05e+04	2.11e+04	1.03e+04	1.18e+02	1.29e+04	5.59e+03	6.29e+03	4.22e+03
F22017	2017	10	1.90e+00	2.48e+02	1.67e+01	2.80e+01	3.92e+01	5.24e-03	1.35e+00	1.37e-01	3.18e-01	3.69e-01	4.16e-05	2.55e-02	4.07e-04	1.30e-03	3.67e-03
F32017	2017	10	5.12e-01	1.84e+02	9.80e+00	3.08e+01	4.48e+01	7.18e-01	7.25e+01	8.59e+00	1.03e+01	1.53e+01	1.10e-03	7.23e+01	4.18e+00	7.43e+00	1.35e+01
F42017	2017	10	3.93e+01	1.47e+02	7.41e+01	8.03e+01	2.41e+01	1.13e+01	7.22e+01	2.76e+01	3.13e+01	1.52e+01	5.98e+00	5.48e+01	1.89e+01	2.02e+01	9.35e+00
F52017	2017	10	5.77e-08	1.50e-03	1.31e-05	8.11e-05	2.17e-04	2.70e-10	3.85e-04	1.28e-06	3.54e-05	8.25e-05	1.14e-13	1.24e-04	9.08e-08	5.90e-06	2.03e-05
F62017	2017	10	1.11e+02	1.58e+03	1.62e+02	2.48e+02	2.68e+02	4.20e+01	9.83e+01	6.04e+01	6.42e+01	1.32e+01	1.71e+01	5.71e+01	3.56e+01	3.57e+01	8.80e+00
F72017	2017	10	2.40e+01	1.22e+02	5.40e+01	5.94e+01	2.25e+01	1.02e+01	4.40e+01	2.80e+01	2.64e+01	8.15e+00	8.00e+00	2.40e+01	1.20e+01	1.28e+01	3.18e+00
F82017	2017	10	1.76e-03	3.62e+00	4.66e-01	7.52e-01	9.93e-01	7.31e-06	6.33e-01	1.05e-04	9.94e-02	1.76e-01	2.44e-08	5.44e-01	2.11e-07	1.42e-02	7.77e-02
F92017	2017	10	9.45e+01	1.56e+03	7.91e+02	7.86e+02	2.85e+02	8.11e+01	9.39e+02	6.27e+02	5.71e+02	2.45e+02	2.57e+01	8.36e+02	3.91e+02	4.05e+02	2.19e+02
F102017	2017	10	1.01e+03	5.43e+04	1.44e+04	1.70e+04	1.23e+04	1.43e+01	8.95e+03	1.29e+03	1.95e+03	2.21e+03	2.56e+00	3.41e+03	2.11e+02	3.54e+02	5.95e+02
F112017	2017	10	3.56e+04	1.12e+07	3.13e+05	1.10e+06	2.20e+06	1.70e+03	1.58e+05	3.61e+04	3.96e+04	2.79e+04	2.10e+03	7.39e+04	3.27e+04	3.08e+04	1.83e+04
F122017	2017	10	6.69e+03	4.43e+05	2.64e+04	1.02e+05	1.40e+05	2.76e+03	6.39e+04	1.34e+04	1.71e+04	1.21e+04	3.14e+02	2.02e+04	4.99e+03	6.59e+03	4.64e+03
F132017	2017	10	7.23e+02	3.73e+04	6.73e+03	9.91e+03	8.99e+03	7.34e+01	1.78e+03	1.57e+02	2.45e+02	2.65e+02	2.34e+01	2.60e+02	4.34e+01	6.68e+01	6.07e+01
F142017	2017	10	1.07e+04	8.20e+05	6.03e+04	1.68e+05	2.03e+05	2.27e+03	3.07e+04	9.76e+03	1.29e+04	8.21e+03	5.03e+01	1.76e+04	4.17e+03	4.28e+03	3.43e+03
F152017	2017	10	1.08e+01	2.94e+04	7.92e+02	2.27e+03	5.32e+03	1.15e+00	1.10e+03	3.63e+02	3.07e+02	2.78e+02	1.27e+00	9.76e+02	2.42e+02	2.56e+02	2.31e+02
F162017	2017	10	4.49e+01	1.38e+03	5.62e+02	5.68e+02	3.25e+02	2.99e+01	1.24e+03	2.54e+02	3.23e+02	3.05e+02	2.45e+01	6.95e+02	4.23e+01	1.72e+02	1.87e+02
F172017	2017	10	3.83e+03	8.55e+04	2.92e+04	3.37e+04	2.01e+04	1.11e+03	4.59e+04	1.42e+04	1.41e+04	9.19e+03	6.35e+01	2.04e+04	1.60e+03	4.41e+03	5.52e+03
F182017	2017	10	8.33e+02	5.39e+04	1.99e+04	2.10e+04	1.27e+04	3.23e+02	2.06e+04	4.49e+03	7.96e+03	6.64e+03	3.20e+01	9.53e+03	1.50e+03	3.02e+03	3.06e+03
F192017	2017	10	4.12e+01	3.36e+02	1.03e+02	1.44e+02	9.53e+01	2.56e+01	2.53e+02	4.46e+01	6.61e+01	5.74e+01	2.22e+01	8.44e+01	3.59e+01	4.76e+01	2.27e+01

func	suite	dim	5k					50k					500k				
			b	w	med	avg	sd	b	w	med	avg	sd	b	w	med	avg	sd
F202017	2017	10	1.17e+02	2.71e+02	1.63e+02	1.71e+02	3.56e+01	1.01e+02	2.46e+02	1.02e+02	1.10e+02	3.31e+01	1.00e+02	2.39e+02	1.00e+02	1.07e+02	2.93e+01
F212017	2017	10	1.02e+02	1.11e+02	1.05e+02	1.06e+02	1.90e+00	1.01e+02	1.13e+02	1.05e+02	1.05e+02	2.44e+00	1.00e+02	1.12e+02	1.05e+02	1.05e+02	2.30e+00
F222017	2017	10	1.43e+02	2.96e+02	1.89e+02	2.03e+02	4.16e+01	1.02e+02	2.04e+02	1.07e+02	1.40e+02	4.67e+01	1.00e+02	2.00e+02	1.01e+02	1.36e+02	4.82e+01
F232017	2017	10	1.34e+02	4.33e+02	3.17e+02	3.14e+02	6.52e+01	1.10e+02	3.42e+02	2.09e+02	2.18e+02	4.50e+01	1.00e+02	3.54e+02	2.01e+02	2.09e+02	4.97e+01
F242017	2017	10	4.80e+02	5.83e+02	4.89e+02	5.02e+02	2.42e+01	4.77e+02	5.68e+02	4.83e+02	4.94e+02	2.10e+01	4.77e+02	5.50e+02	4.82e+02	4.92e+02	1.65e+01
F252017	2017	10	8.40e-01	5.42e+02	4.67e+02	4.48e+02	9.18e+01	6.84e-02	4.68e+02	4.67e+02	4.10e+02	1.51e+02	6.04e-03	4.68e+02	4.67e+02	3.56e+02	1.99e+02
F262017	2017	10	3.91e+02	4.04e+02	3.96e+02	3.96e+02	3.50e+00	3.90e+02	4.03e+02	3.95e+02	3.95e+02	3.40e+00	3.90e+02	3.98e+02	3.92e+02	3.93e+02	3.05e+00
F272017	2017	10	2.95e+01	4.93e+02	7.94e+01	2.24e+02	1.90e+02	1.32e+00	3.73e+02	5.59e+00	4.56e+01	1.06e+02	8.73e-02	1.97e+01	1.92e-01	2.01e+00	5.14e+00
F282017	2017	10	1.01e+03	3.83e+04	3.97e+03	8.25e+03	9.86e+03	9.97e+02	2.35e+04	1.41e+03	3.07e+03	4.75e+03	9.84e+02	1.65e+04	1.07e+03	1.69e+03	2.35e+03
F292017	2017	10	1.23e+04	3.13e+07	3.63e+04	1.06e+06	4.54e+06	3.71e+03	2.17e+05	2.30e+04	3.28e+04	3.68e+04	5.10e+03	5.15e+04	2.15e+04	2.44e+04	1.17e+04
F12022	2022	10	9.11e-01	1.10e+02	1.68e+01	3.06e+01	3.15e+01	2.09e-03	5.20e-02	8.30e-03	1.14e-02	1.10e-02	6.99e-06	2.34e-04	4.52e-05	5.76e-05	4.46e-05
F22022	2022	10	7.00e+00	9.09e+01	9.08e+00	2.12e+01	2.74e+01	4.57e+00	8.88e+01	8.92e+00	1.44e+01	2.01e+01	9.51e-02	7.10e+01	6.61e+00	8.13e+00	9.27e+00
F32022	2022	10	2.53e-05	2.18e-03	1.52e-04	3.57e-04	5.11e-04	1.76e-07	5.75e-06	6.55e-07	8.87e-07	8.35e-07	1.31e-09	9.29e-09	4.26e-09	4.62e-09	2.06e-09
F42022	2022	10	2.20e+01	1.22e+02	6.02e+01	6.62e+01	2.61e+01	1.20e+01	8.80e+01	4.60e+01	4.76e+01	1.69e+01	8.00e+00	4.00e+01	2.40e+01	2.42e+01	6.28e+00
F52022	2022	10	4.02e-04	3.20e+00	5.54e-01	7.67e-01	7.85e-01	2.90e-06	1.18e+00	8.96e-02	2.19e-01	3.01e-01	2.23e-08	5.44e-01	2.78e-07	3.54e-02	1.01e-01
F62022	2022	10	2.80e+04	1.29e+05	6.42e+04	6.67e+04	2.09e+04	2.10e+04	9.97e+04	5.51e+04	5.47e+04	1.81e+04	1.01e+04	6.81e+04	3.99e+04	3.51e+04	1.33e+04
F72022	2022	10	2.72e+01	1.53e+02	3.46e+01	6.19e+01	4.70e+01	1.84e+01	1.32e+02	2.38e+01	3.20e+01	2.81e+01	7.24e-01	2.56e+01	2.14e+01	2.05e+01	4.07e+00
F82022	2022	10	2.59e+01	1.99e+03	5.74e+01	3.06e+02	5.38e+02	5.52e+00	4.32e+02	2.45e+01	3.77e+01	6.12e+01	2.09e+01	2.58e+01	2.26e+01	2.27e+01	9.95e-01
F92022	2022	10	3.60e+02	4.45e+02	3.70e+02	3.74e+02	1.65e+01	3.59e+02	3.69e+02	3.65e+02	3.64e+02	3.74e+00	4.87e-03	3.59e+02	1.31e-02	1.34e+02	1.75e+02
F102022	2022	10	2.21e+02	9.69e+02	2.27e+02	2.83e+02	1.27e+02	2.21e+02	3.53e+02	2.23e+02	2.40e+02	4.10e+01	2.19e+02	3.39e+02	2.23e+02	2.28e+02	2.12e+01
F112022	2022	10	3.76e-01	8.77e+02	2.33e+00	1.59e+02	3.35e+02	2.23e-02	8.76e+02	4.37e-02	1.81e+01	1.23e+02	2.16e-03	7.71e-03	3.62e-03	3.60e-03	9.85e-04
F122022	2022	10	1.64e+02	1.69e+02	1.67e+02	1.67e+02	1.01e+00	1.64e+02	1.67e+02	1.66e+02	1.66e+02	1.04e+00	1.64e+02	1.67e+02	1.66e+02	1.66e+02	1.06e+00

WOA — estadisticas por funcion, D = 50

error al optimo sobre 51 corridas · b=best w=worst med=median avg=mean sd=std

func	suite	dim	5k					50k					500k				
			b	w	med	avg	sd	b	w	med	avg	sd	b	w	med	avg	sd
F12014	2014	50	1.43e+08	7.31e+08	2.86e+08	3.15e+08	1.15e+08	3.01e+07	1.39e+08	6.13e+07	6.49e+07	2.37e+07	9.68e+06	3.89e+07	2.08e+07	2.14e+07	7.77e+06
F22014	2014	50	1.26e+10	3.76e+10	2.17e+10	2.25e+10	4.95e+09	2.06e+08	3.57e+09	7.46e+08	8.81e+08	6.35e+08	1.39e+06	2.34e+08	2.78e+07	4.82e+07	5.08e+07
F32014	2014	50	1.45e+05	3.85e+05	2.52e+05	2.54e+05	5.40e+04	2.65e+04	7.30e+04	4.55e+04	4.55e+04	1.01e+04	1.75e+03	1.71e+04	6.96e+03	6.96e+03	2.93e+03
F42014	2014	50	8.78e+02	3.42e+03	1.76e+03	1.86e+03	5.57e+02	1.16e+02	5.14e+02	2.92e+02	2.88e+02	1.08e+02	9.85e+01	3.35e+02	1.64e+02	1.66e+02	5.98e+01
F52014	2014	50	2.12e+01	2.14e+01	2.13e+01	2.13e+01	4.18e-02	2.07e+01	2.11e+01	2.10e+01	2.10e+01	1.01e-01	2.02e+01	2.06e+01	2.04e+01	2.04e+01	9.47e-02
F62014	2014	50	4.51e+01	6.60e+01	5.63e+01	5.64e+01	3.95e+00	3.47e+01	5.67e+01	4.80e+01	4.73e+01	4.88e+00	3.71e+01	5.48e+01	4.38e+01	4.47e+01	4.54e+00
F72014	2014	50	1.02e+02	3.93e+02	1.98e+02	2.13e+02	5.70e+01	3.16e+00	3.02e+01	7.55e+00	8.27e+00	4.53e+00	8.98e-01	4.05e+00	1.22e+00	1.38e+00	5.88e-01
F82014	2014	50	3.19e+02	5.57e+02	4.36e+02	4.39e+02	5.24e+01	1.75e+02	3.59e+02	2.50e+02	2.55e+02	4.39e+01	1.09e+02	2.13e+02	1.49e+02	1.50e+02	2.55e+01
F92014	2014	50	4.02e+02	6.86e+02	5.35e+02	5.33e+02	7.72e+01	2.88e+02	5.93e+02	3.95e+02	4.11e+02	8.31e+01	2.38e+02	6.36e+02	4.16e+02	4.18e+02	8.05e+01
F102014	2014	50	7.84e+03	1.21e+04	1.03e+04	1.02e+04	9.32e+02	5.03e+03	7.86e+03	6.80e+03	6.66e+03	7.57e+02	2.55e+03	6.30e+03	4.37e+03	4.33e+03	9.28e+02
F112014	2014	50	9.70e+03	1.29e+04	1.14e+04	1.15e+04	7.45e+02	5.85e+03	1.02e+04	7.89e+03	7.80e+03	8.81e+02	4.86e+03	8.74e+03	6.95e+03	6.95e+03	8.81e+02
F122014	2014	50	2.06e+00	5.18e+00	3.67e+00	3.63e+00	6.97e-01	7.66e-01	2.53e+00	1.31e+00	1.35e+00	3.81e-01	2.02e-01	1.19e+00	4.97e-01	5.62e-01	2.26e-01
F132014	2014	50	8.29e-01	3.96e+00	2.63e+00	2.39e+00	8.15e-01	6.23e-01	1.14e+00	8.22e-01	8.54e-01	1.38e-01	5.16e-01	1.03e+00	7.81e-01	7.78e-01	1.23e-01
F142014	2014	50	2.47e+01	8.77e+01	5.85e+01	5.86e+01	1.25e+01	2.82e-01	1.46e+00	1.13e+00	1.01e+00	3.21e-01	3.21e-01	1.33e+00	1.16e+00	1.11e+00	2.13e-01
F152014	2014	50	9.10e+03	6.37e+05	7.91e+04	1.18e+05	1.30e+05	3.60e+01	1.38e+02	5.76e+01	6.71e+01	2.49e+01	1.89e+01	4.39e+01	2.82e+01	2.96e+01	5.71e+00
F162014	2014	50	2.25e+01	2.36e+01	2.31e+01	2.30e+01	2.78e-01	2.09e+01	2.30e+01	2.21e+01	2.21e+01	5.11e-01	2.04e+01	2.23e+01	2.16e+01	2.15e+01	4.72e-01
F172014	2014	50	4.60e+06	7.71e+07	1.73e+07	1.98e+07	1.21e+07	1.28e+06	1.39e+07	4.40e+06	4.86e+06	2.47e+06	4.87e+05	3.70e+06	1.47e+06	1.64e+06	7.95e+05
F182014	2014	50	9.70e+06	1.02e+08	3.45e+07	3.81e+07	2.20e+07	3.45e+05	7.99e+05	5.20e+05	5.46e+05	1.16e+05	1.94e+05	4.60e+05	3.57e+05	3.52e+05	6.00e+04
F192014	2014	50	1.22e+04	4.19e+06	2.42e+05	6.62e+05	9.69e+05	9.83e+01	4.26e+05	5.64e+03	3.57e+04	8.77e+04	4.76e+01	1.60e+04	1.45e+02	2.02e+03	3.87e+03
F202014	2014	50	1.99e+06	3.08e+10	1.57e+09	4.58e+09	6.90e+09	3.20e+05	7.22e+05	4.99e+05	4.92e+05	8.93e+04	2.59e+05	6.53e+05	4.03e+05	4.10e+05	8.98e+04
F212014	2014	50	6.44e+06	1.59e+08	2.64e+07	3.65e+07	2.93e+07	8.18e+05	9.80e+06	3.52e+06	3.84e+06	2.12e+06	5.13e+05	4.25e+06	1.51e+06	1.68e+06	8.85e+05
F222014	2014	50	4.84e+03	1.13e+06	9.62e+03	5.32e+04	1.68e+05	3.74e+03	1.12e+04	6.35e+03	6.76e+03	1.79e+03	5.92e+02	4.02e+03	2.52e+03	2.58e+03	7.83e+02
F232014	2014	50	2.93e+02	3.75e+02	3.29e+02	3.28e+02	1.57e+01	2.36e+02	2.38e+02	2.36e+02	2.36e+02	3.42e-01	2.35e+02	2.36e+02	2.35e+02	2.35e+02	5.10e-02
F242014	2014	50	3.82e+02	4.71e+02	4.27e+02	4.26e+02	2.02e+01	2.41e+02	3.06e+02	2.75e+02	2.79e+02	1.32e+01	2.00e+02	2.96e+02	2.67e+02	2.60e+02	2.69e+01
F252014	2014	50	2.28e+02	3.05e+02	2.67e+02	2.66e+02	1.77e+01	2.14e+02	2.53e+02	2.31e+02	2.31e+02	8.38e+00	2.09e+02	2.39e+02	2.18e+02	2.19e+02	6.39e+00
F262014	2014	50	1.01e+02	4.63e+02	1.03e+02	1.83e+02	1.38e+02	1.01e+02	4.44e+02	1.01e+02	1.31e+02	9.02e+01	1.01e+02	4.26e+02	1.01e+02	1.07e+02	4.55e+01
F272014	2014	50	1.53e+03	2.10e+03	1.84e+03	1.82e+03	1.33e+02	1.29e+03	1.99e+03	1.59e+03	1.60e+03	1.36e+02	1.23e+03	1.83e+03	1.50e+03	1.52e+03	1.36e+02
F282014	2014	50	1.58e+03	6.06e+03	2.82e+03	2.92e+03	9.33e+02	1.33e+03	3.42e+03	2.34e+03	2.24e+03	5.87e+02	1.32e+03	3.69e+03	2.11e+03	2.12e+03	5.84e+02
F292014	2014	50	3.38e+07	7.03e+08	3.59e+08	3.71e+08	1.47e+08	1.79e+05	1.55e+08	5.91e+07	5.99e+07	4.92e+07	1.68e+04	9.27e+07	7.26e+05	1.83e+07	2.43e+07
F302014	2014	50	9.90e+07	5.83e+09	1.39e+09	1.70e+09	1.27e+09	2.84e+06	6.92e+07	1.83e+07	2.32e+07	1.64e+07	1.50e+05	1.72e+07	2.30e+06	2.88e+06	2.78e+06
F12017	2017	50	1.16e+10	3.53e+10	2.17e+10	2.15e+10	4.70e+09	1.27e+08	2.19e+09	4.31e+08	5.17e+08	4.03e+08	2.45e+06	8.00e+07	2.99e+07	2.89e+07	1.74e+07
F22017	2017	50	3.86e+04	1.11e+05	7.57e+04	7.56e+04	1.59e+04	7.69e+02	1.25e+04	2.73e+03	3.58e+03	2.38e+03	3.74e+00	1.22e+03	1.07e+02	1.61e+02	1.89e+02
F32017	2017	50	8.42e+02	3.24e+03	1.62e+03	1.71e+03	5.82e+02	1.66e+02	3.67e+02	2.41e+02	2.49e+02	4.68e+01	7.45e+01	2.63e+02	1.64e+02	1.64e+02	2.78e+01
F42017	2017	50	1.22e+04	3.13e+04	2.07e+04	2.12e+04	4.79e+03	7.05e+02	3.58e+03	1.27e+03	1.40e+03	5.24e+02	3.59e+02	1.19e+03	5.64e+02	5.92e+02	1.51e+02
F52017	2017	50	3.99e-03	3.70e-02	1.77e-02	1.84e-02	6.36e-03	8.24e-05	2.05e-02	2.88e-03	3.53e-03	3.22e-03	7.75e-05	1.06e-02	1.40e-03	2.11e-03	2.16e-03
F62017	2017	50	4.78e+05	1.05e+06	7.07e+05	7.13e+05	1.46e+05	8.35e+03	7.23e+04	3.24e+04	3.32e+04	1.65e+04	1.08e+03	8.19e+03	2.91e+03	3.64e+03	2.22e+03
F72017	2017	50	7.53e+02	1.51e+03	1.08e+03	1.11e+03	1.82e+02	4.49e+02	1.19e+03	8.55e+02	8.32e+02	1.85e+02	4.74e+02	1.18e+03	9.20e+02	9.09e+02	1.66e+02
F82017	2017	50	3.21e+01	1.07e+02	6.44e+01	6.66e+01	1.86e+01	2.41e+01	7.78e+01	4.62e+01	4.85e+01	1.19e+01	2.23e+01	7.32e+01	4.45e+01	4.51e+01	1.22e+01
F92017	2017	50	9.86e+03	1.29e+04	1.18e+04	1.16e+04	8.09e+02	5.80e+03	1.10e+04	8.31e+03	8.22e+03	1.10e+03	4.80e+03	9.79e+03	7.13e+03	7.16e+03	9.32e+02
F102017	2017	50	1.08e+06	3.83e+07	9.57e+06	1.13e+07	9.26e+06	1.18e+05	6.36e+05	3.25e+05	3.50e+05	1.07e+05	1.74e+04	1.94e+05	8.94e+04	9.08e+04	3.74e+04
F112017	2017	50	3.77e+08	1.71e+09	7.52e+08	8.52e+08	3.34e+08	7.44e+06	1.57e+08	4.55e+07	5.52e+07	3.68e+07	1.27e+06	5.04e+07	3.85e+06	6.40e+06	7.59e+06
F122017	2017	50	3.00e+08	3.08e+09	1.05e+09	1.10e+09	5.99e+08	1.64e+06	5.97e+07	1.43e+07	1.98e+07	1.58e+07	3.35e+05	2.17e+07	1.41e+06	2.33e+06	3.16e+06
F132017	2017	50	6.14e+05	1.99e+07	4.33e+06	5.22e+06	3.89e+06	4.25e+05	5.04e+06	1.56e+06	1.79e+06	1.04e+06	3.02e+05	1.89e+06	8.01e+05	8.47e+05	4.13e+05
F142017	2017	50	1.35e+07	4.17e+08	5.82e+07	9.29e+07	8.40e+07	2.60e+05	5.56e+06	8.66e+05	1.17e+06	8.94e+05	2.35e+05	1.20e+06	3.19e+05	4.02e+05	2.21e+05
F152017	2017	50	2.42e+05	9.63e+06	8.88e+05	1.45e+06	1.68e+06	3.56e+04	6.09e+05	1.46e+05	2.27e+05	1.73e+05	2.95e+03	4.76e+04	1.55e+04	1.74e+04	9.87e+03
F162017	2017	50	6.10e+04	2.08e+06	1.39e+05	1.99e+05	2.90e+05	5.09e+03	5.44e+04	1.69e+04	1.88e+04	9.31e+03	1.89e+03	1.05e+04	2.94e+03	3.51e+03	1.81e+03
F172017	2017	50	1.47e+06	4.81e+07	1.25e+07	1.43e+07	1.04e+07	8.05e+05	1.61e+07	1.52e+06	2.13e+06	2.32e+06	4.12e+05	1.31e+06	6.96e+05	7.25e+05	1.83e+05
F182017	2017	50	3.63e+09	1.11e+11	2.55e+10	3.13e+10	2.35e+10	1.89e+05	4.22e+08	1.98e+06	2.10e+07	7.74e+07	8.30e+04	5.54e+05	2.67e+05	2.74e+05	1.07e+05
F192017	2017	50	2.36e+03	7.45e+03	4.44e+03	4.46e+03	1.25e+03	4.34e+02	3.32e+03	1.46e+03	1.56e+03	6.61e+02	1.42e+02	2.24e+03	8.81e+02	9.54e+02	5.04e+02

func	suite	dim	5k					50k					500k				
			b	w	med	avg	sd	b	w	med	avg	sd	b	w	med	avg	sd
F202017	2017	50	2.47e+03	3.42e+04	1.75e+04	1.50e+04	9.29e+03	8.44e+02	3.69e+03	1.57e+03	1.66e+03	6.50e+02	2.01e+02	1.08e+03	6.90e+02	6.70e+02	2.21e+02
F212017	2017	50	3.97e+02	1.21e+04	4.61e+02	1.52e+03	3.24e+03	2.37e+02	7.21e+03	2.79e+02	6.32e+02	1.43e+03	2.08e+02	5.96e+03	2.43e+02	5.16e+02	1.13e+03
F222017	2017	50	1.79e+04	3.57e+04	2.25e+04	2.33e+04	4.11e+03	1.10e+03	9.99e+03	2.73e+03	3.14e+03	1.59e+03	3.46e+02	2.63e+03	9.04e+02	9.76e+02	5.42e+02
F232017	2017	50	9.52e+03	2.38e+04	1.22e+04	1.29e+04	2.69e+03	9.66e+02	3.41e+03	2.04e+03	2.09e+03	6.62e+02	3.11e+02	1.79e+03	5.77e+02	6.54e+02	3.59e+02
F242017	2017	50	1.26e+03	3.81e+03	2.32e+03	2.40e+03	5.19e+02	8.42e+02	1.48e+03	1.05e+03	1.07e+03	1.50e+02	7.25e+02	1.27e+03	8.56e+02	8.74e+02	1.15e+02
F252017	2017	50	1.66e+03	2.67e+03	1.82e+03	1.83e+03	1.54e+02	1.44e+03	1.74e+03	1.62e+03	1.60e+03	7.61e+01	1.22e+03	1.63e+03	1.58e+03	1.45e+03	1.63e+02
F262017	2017	50	6.20e+02	1.03e+03	7.44e+02	7.57e+02	8.67e+01	5.47e+02	9.32e+02	6.23e+02	6.30e+02	6.06e+01	5.18e+02	7.70e+02	6.08e+02	6.16e+02	5.58e+01
F272017	2017	50	6.32e+02	1.02e+03	7.64e+02	7.76e+02	9.08e+01	4.17e+02	5.25e+02	4.86e+02	4.84e+02	1.96e+01	1.85e+02	4.96e+02	4.52e+02	3.99e+02	9.30e+01
F282017	2017	50	2.11e+08	8.77e+09	1.60e+09	2.25e+09	1.96e+09	2.31e+05	8.95e+07	3.23e+06	8.83e+06	1.51e+07	9.26e+03	7.90e+06	6.86e+04	5.05e+05	1.42e+06
F292017	2017	50	1.81e+09	3.84e+10	8.67e+09	1.15e+10	7.52e+09	4.77e+06	4.75e+08	7.98e+07	1.01e+08	9.59e+07	1.38e+06	6.81e+07	2.19e+07	2.91e+07	1.93e+07
F12022	2022	20	3.27e+02	4.75e+03	1.35e+03	1.62e+03	9.52e+02	4.82e-01	8.30e+01	8.31e+00	1.86e+01	2.35e+01	1.81e-03	4.99e-02	7.67e-03	9.97e-03	8.80e-03
F22022	2022	20	5.65e+01	3.14e+02	9.86e+01	1.16e+02	6.14e+01	3.05e+01	1.98e+02	5.21e+01	6.42e+01	3.09e+01	6.52e+00	1.73e+02	4.91e+01	5.09e+01	2.63e+01
F32022	2022	20	3.46e-03	3.78e-02	1.04e-02	1.25e-02	7.01e-03	1.32e-05	1.69e-03	4.63e-05	1.19e-04	3.11e-04	1.27e-07	2.28e-06	3.14e-07	4.08e-07	3.33e-07
F42022	2022	20	8.87e+01	3.63e+02	2.33e+02	2.23e+02	6.52e+01	1.06e+02	3.54e+02	2.08e+02	2.09e+02	6.12e+01	4.60e+01	2.42e+02	1.64e+02	1.68e+02	4.31e+01
F52022	2022	20	1.51e+00	1.63e+01	7.98e+00	7.97e+00	3.50e+00	5.95e-01	1.60e+01	6.89e+00	6.87e+00	3.76e+00	9.92e-01	1.26e+01	4.73e+00	4.91e+00	2.90e+00
F62022	2022	20	3.15e+05	3.46e+06	8.74e+05	1.06e+06	6.88e+05	3.98e+04	1.77e+05	1.17e+05	1.16e+05	3.26e+04	3.50e+04	1.93e+05	1.16e+05	1.11e+05	3.30e+04
F72022	2022	20	7.55e+01	1.28e+03	2.14e+02	2.86e+02	2.32e+02	3.88e+01	3.81e+02	6.95e+01	1.03e+02	7.66e+01	2.66e+01	5.87e+02	4.47e+01	6.53e+01	7.90e+01
F82022	2022	20	9.36e+02	1.05e+04	3.95e+03	4.19e+03	2.37e+03	4.53e+01	3.60e+03	6.82e+02	7.73e+02	6.68e+02	2.51e+01	5.28e+02	3.94e+01	9.26e+01	1.06e+02
F92022	2022	20	3.44e+02	5.17e+02	3.71e+02	3.80e+02	3.50e+01	3.36e+02	3.55e+02	3.39e+02	3.40e+02	3.99e+00	3.36e+02	3.36e+02	3.36e+02	3.36e+02	2.82e-02
F102022	2022	20	4.11e+02	3.92e+03	2.67e+03	2.33e+03	1.16e+03	3.99e+02	3.34e+03	1.97e+03	1.70e+03	1.00e+03	3.95e+02	3.05e+03	4.11e+02	1.04e+03	8.89e+02
F112022	2022	20	9.28e+00	1.58e+03	3.07e+01	4.81e+02	6.68e+02	1.43e-01	2.45e+01	2.93e+00	7.84e+00	8.63e+00	1.27e-02	2.28e+01	2.81e-02	4.19e+00	7.16e+00
F122022	2022	20	2.40e+02	3.42e+02	2.61e+02	2.66e+02	2.11e+01	2.39e+02	2.87e+02	2.57e+02	2.57e+02	1.28e+01	2.38e+02	2.84e+02	2.53e+02	2.55e+02	1.31e+01